# JEFFREY DAI

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## EDUCATION

**McGill University** Montreal, QC
*Bachelor of Engineering, Computer Engineering* *Expected April 2029*
**CGPA: 3.89/4.00**

## EXPERIENCE

**Information Technology Intern** May 2026 – Present
*Canada Revenue Agency, Enterprise Fraud Management, IT Branch*
Continuing part-time (20 hours/week) after a four-month internship.

- Built and maintained Java and XML information mappings for enterprise fraud-management systems.
- Develop remotely through SSH in VS Code, using Git for version control, Azure DevOps for pull requests and pipelines, and Jira for task tracking.
- Supported monitoring of database queries, fraud-detection services, and JSON-based message queues.

## PROJECTS

**Chinatown Online** | *React, Node.js, Express, Socket.IO, Vite, Render* GitHub | Live Demo
**Demo password:** 2026CSUroom

- Built and deployed a server-authoritative multiplayer board game for 3–5 players, with room codes and private player state.
- Implemented live trading and reconnect handling; added automated end-to-end smoke tests.

**Dishly Recipe Match** | *React, Node.js, Express, Gemini, Spoonacular, Playwright* GitHub | Website

- Built a recipe-matching app that translates natural-language cravings into dietary and nutrition filters and presents normalized results in a responsive, swipeable interface.
- Implemented API routes, recipe integrations, session-based deck persistence, and recipe detail views.
- Added automated unit, API, and end-to-end tests during a 36-hour CUhacking project.

**Claw Machine** | *Arduino, Stepper Motors, 3D Printing* Personal Project

- Designed and built a claw machine with a 3D-printer-style gantry controlled by two Arduinos.
- Wrote code to drive three stepper motors: two for gantry movement and one for the claw.

## ADDITIONAL HACKATHONS

**Ignition Hacks:** EduAtlas (Java education game) | **Hack the Hill II:** Serenity (Arduino health monitor)
**Counterspell:** PvP game | **Code Jam 15:** In Class Bored Games (minigame app)

## LEADERSHIP & ACTIVITIES

**Experience Team Lead** 2025 – Present
*McHacks*

- Lead a team supporting a hackathon with 700 participants selected from over 1,500 applications.
- Organize backend, frontend, and AI chatbot workshops; coordinate sessions with external partners.

## TECHNICAL SKILLS

**Languages:** JavaScript, Java, Python, C, C++, Bash, VHDL
**Web & Testing:** React, Node.js, Express, Socket.IO, Playwright
**Developer Tools:** Git, Azure DevOps, Jira, VS Code, SSH, Claude, Render
**Embedded & Hardware:** Arduino, stepper motors, PCB design, 3D printing
**APIs & Data:** Gemini API, Spoonacular API, XML, JSON
**Spoken Languages:** English (fluent), French (advanced)